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$$\begin{aligned}\int \cos^4 x dx &= \int (\cos^2 x)^2 dx = \int \left(\frac{1 + \cos 2x}{2} \right) dx \\&= \frac{1}{4} \int (1 + 2 \cos 2x + \cos^2 2x) dx = \frac{1}{4} x + \frac{1}{4} \sin 2x + \frac{1}{4} \int \left(\frac{1 + \cos 4x}{2} \right) dx \\&= \frac{1}{4} x + \frac{1}{4} \sin 2x + \frac{1}{8} \int (1 + \cos 4x) dx = \frac{1}{4} x + \frac{1}{4} \sin 2x + \frac{1}{8} x + \frac{1}{32} \sin 4x \\&= \frac{3}{8} x + \frac{1}{4} \sin 2x + \frac{1}{32} \sin 4x + c\end{aligned}$$

References